

STUDENT PROJECT THEMES

Korea AI in Action

Think AI. Create Impact.

Proposed project themes for the final-day group work — in collaboration with KFAS.



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Project Approach & Deliverables

Your approach

- 1** Identify a real-world challenge within your theme
- 2** Develop a clear, solution-oriented concept
- 3** Explore it with simplified tools, scenarios, or datasets
- 4** Present your work in a structured, visual format

What to deliver

-  **Concept Proposal**
Problem statement and proposed solution, with simple diagrams or logic
-  **Basic Analysis or Insight**
Simple interpretation of data or scenarios, where applicable
-  **Prototype, Model or Simulation**
A physical model, digital simulation, or mock-up / workflow
-  **Final Presentation**
Problem, approach, solution, and expected impact

Student Project Themes

Think AI. Create Impact.

Proposed by KFAS — apply what you've learned to a real-world challenge.



Energy Efficiency in Buildings

Reduce energy use and optimize cooling for hot climates like Kuwait — smarter energy management and building practices.

e.g., building-energy datasets · Kuwait MEW conservation code



Smart Farming & Food Security

Support sustainable agriculture — smart irrigation, environmental monitoring, even home-scale growing systems.

e.g., soil & irrigation data · crop-recommendation datasets



Intelligent Healthcare

Apply data and AI to healthcare — early-detection concepts and more efficient, accessible systems.

e.g., public health indicators · diabetes dataset

Example datasets and references are provided to guide exploration and may be adapted to student level and resources.

Energy Efficiency in Buildings



Improve energy efficiency and cut consumption in buildings — especially in high-temperature environments like Kuwait. The focus is on practical, solution-oriented ideas rather than technical implementation, using Kuwait's Energy Conservation Code as a baseline reference.

Explore with AI

- Optimize cooling systems for hot climates
- Design smart energy-management solutions
- Improve existing building practices
- Analyze consumption patterns and propose optimizations



Example datasets

- World Energy Consumption — global use & sources
- Energy Efficiency — building design & heating/cooling loads
- Global Weather — temperature & climate



Key references

- World Bank Open Data — energy indicators
- UNEP — environmental data & reports
- UN Data Portal — country statistics
- Paris Agreement — climate framework
- Kuwait MEW Energy Conservation Code (R-6/2018)
- Kuwait Meteorological Department
- Kaggle — building-energy datasets

Smart Farming & Food Security



Explore how AI can support sustainable agriculture and food security, especially in resource-constrained environments. The scope can extend to small-scale or home-based smart farming, such as indoor growing and controlled environments, making it practical and relatable.

Explore with AI

- Smart irrigation systems
- Environmental monitoring
- Data-informed crop-management decisions
- Home-scale / indoor growing systems
- Predictive models for smarter resource use



Example datasets

- Crop Recommendation — soil, climate & optimal crops
- Global Weather — temperature & climate
- Soil moisture & irrigation datasets



Key references

- FAO (FAOSTAT) — agriculture & food security
- World Bank Open Data — agriculture & water use
- UNEP — environmental & sustainability data
- UN Data Portal — country-level indicators
- Kaggle — smart-farming datasets

Intelligent Healthcare



Use data and AI to support improved healthcare outcomes. Students explore basic data interpretation, early-detection concepts, and digital solutions that make healthcare more efficient and accessible — developing simple, practical ideas with structured guidance.

Explore with AI

- Basic health-data interpretation
- Early-detection concepts
- Digital solutions for efficiency & accessibility
- Support decision-making in health systems



Example datasets

- Diabetes — health indicators for at-risk individuals
- Anonymized public health-statistics datasets



Key references

- WHO — global health data & indicators
- World Bank Open Data — healthcare indicators
- UN Data Portal — country-level health statistics
- Kaggle — healthcare datasets (educational use)